of colour variation. This is due to the fact that the genes are segregating and recombination takes place. Another evidence for the absence of blending inheritance is that the parental phenotypes dark red and white appear again in F_2. There is no blending of genes, only the phenotype. The cumulative effect of several pairs of gene interaction gives rise to many shades of kernel colour. He hypothesized that the two loci must contribute additively to the kernel colour of wheat. The contribution of each red allele to the kernel colour of wheat is additive.

2.7 Extra Chromosomal Inheritance or Extra Nuclear Inheritance (Cytoplasmic Inheritance)

DNA is the universal genetic material. Genes located in nuclear chromosomes follow Mendelian inheritance. But certain traits are governed either by the chloroplast or mitochondrial genes. This phenomenon is known as extra nuclear inheritance. It is a kind of Non-Mendelian inheritance. Since it involves cytoplasmic organelles such as chloroplast and mitochondrion that act as inheritance vectors, it is also called Cytoplasmic inheritance. It is based on independent, self-replicating extra chromosomal unit called plasmogene located in the cytoplasmic organelles, chloroplast and mitochondrion.

Chloroplast Inheritance

It is found in 4 O' Clock plant (*Mirabilis jalapa*). In this, there are two types of variegated leaves namely dark green leaved plants and pale green leaved plants. When the pollen of dark green leaved plant (male) is transferred to the stigma of pale green leaved plant (female) and pollen of pale green leaved plant is transferred to the stigma of dark green leaved plant, the F_1 generation of both the crosses must be identical as per Mendelian inheritance. But in the reciprocal cross the F_1 plant differs from each other. In each cross, the F_1 plant reveals the character of the plant which is used as female plant.

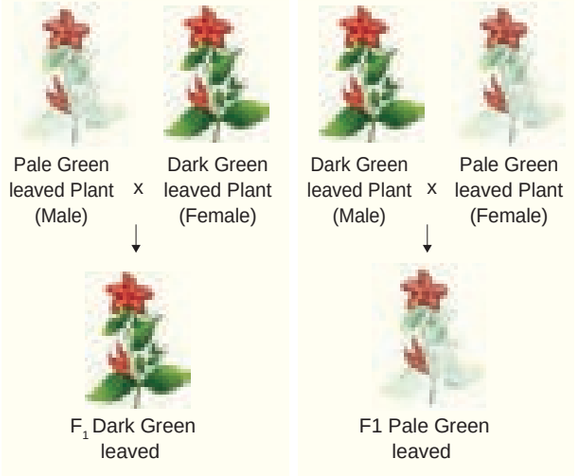


Figure 2.19: Chloroplast inheritance

This inheritance is not through nuclear gene. It is due to the chloroplast gene found in the ovum of the female plant which contributes the cytoplasm during fertilization since the male gamete contribute only the nucleus but not cytoplasm.

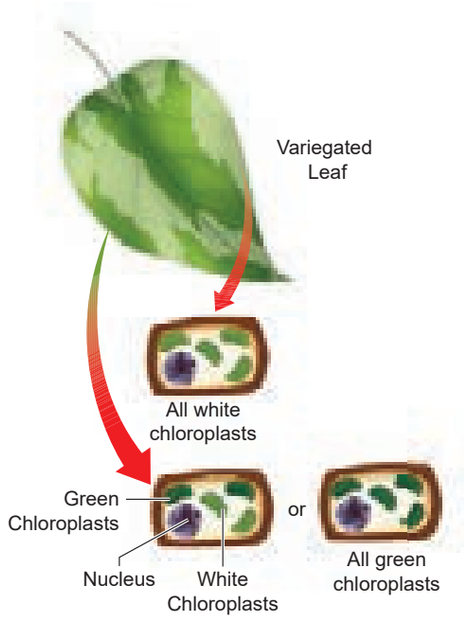


Figure 2.21: A cellular explanation of the variegated phenotype of the leaves in *Mirabilis jalapa*

Recently it has been discovered that cytoplasmic genetic male sterility is common in many plant species. This sterility is maintained by the influence of both nuclear and cytoplasmic genes. There are commonly two types of cytoplasm N (normal) and S (sterile). The genes

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